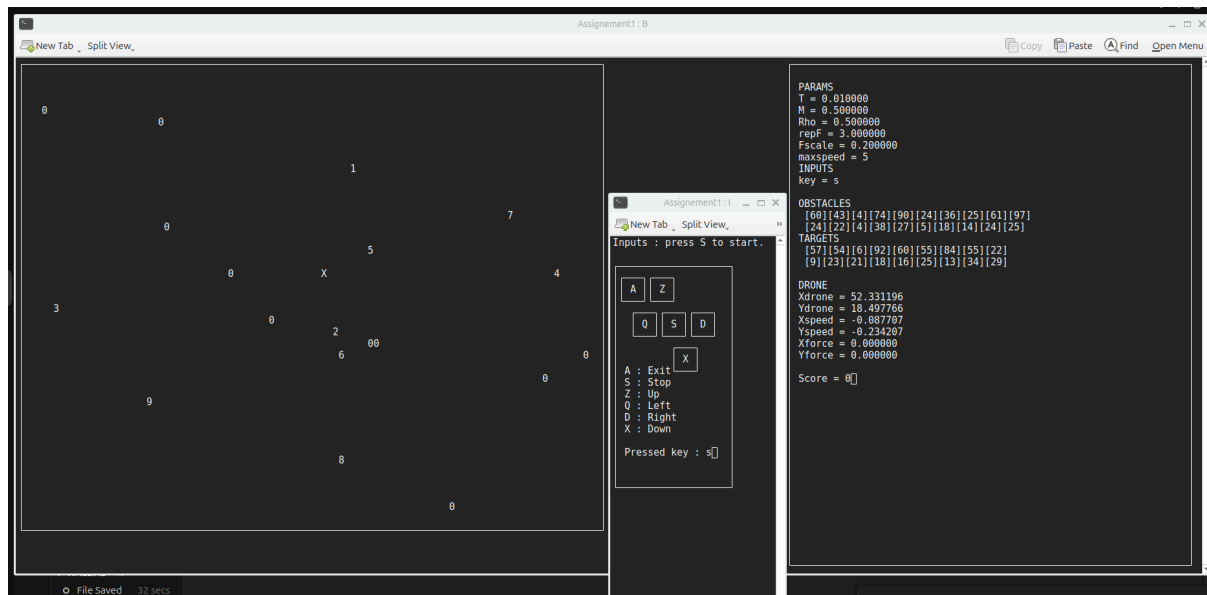


DRONE SIMULATOR PROJECT

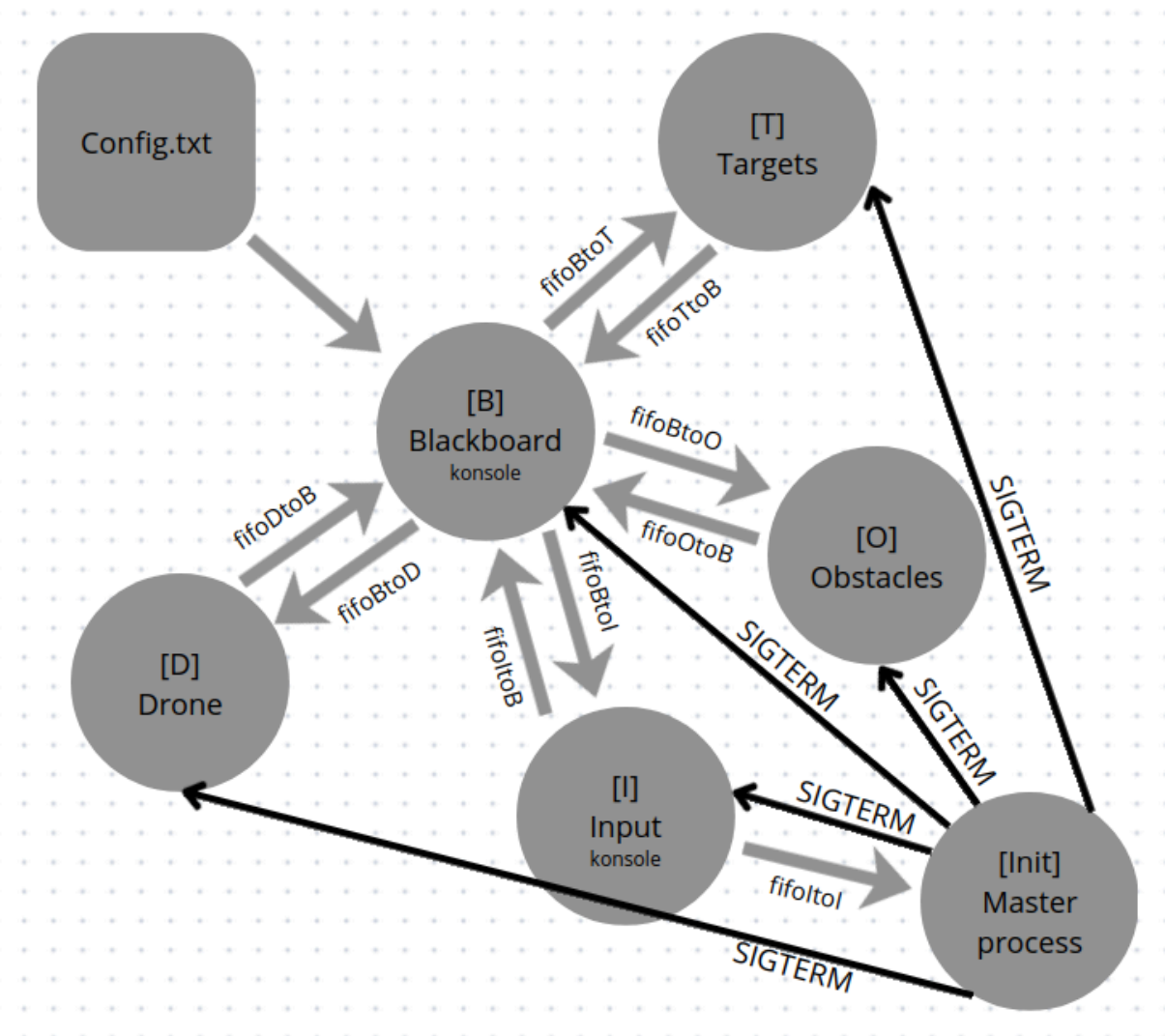
Assignment 1



Project members :

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1. sketch of the architecture



2. short definitions of all active components :

Master process [Init] :

- create all the fifos of this project
- use the functions `fork()+exec()` to launch the 5 other programs
- wait a message from [I] when the user want to quit
- send a signal to all the other programs to terminate
- wait for all the programs to terminate
- terminate cleanly

Blackboard [B] : Store all the values of the project and communicate it to the different programs.

- endless loop, except if received the terminate signal
- read the content of the file at each iteration and update the values of variables
- use the select() function with all the input fifos
- use a special two-pipe protocol to communicate with the programs
- can send or receive data
- calculate the score of the player
- use ncurses to print the scene on the screen and additional informations
- terminate cleanly

Drone [D] : Update the values of the drone representation with interactive dynamic equations.

- endless loop, except if received the terminate signal
- interaction period can be changed through config file
- get value from the blackboard
- use dynamic equations to calculate new values
- send the new values to the blackboard
- terminate cleanly

Input [I] : Read the key pressed.

- endless loop, except if received the terminate signal
- use ncurses to read the key pressed
- the key must be one of the keys selected for this project (a,z,q,s,..)
- send the key value to the blackboard
- if the key is the one to terminate, send a message to the master process
- terminate cleanly

Obstacles [O] : Generate obstacles.

- endless loop, except if received the terminate signal
- long period
- generate new coordinates for the obstacles with random values from /dev/urandom
- send the coordinates to the blackboard
- terminate cleanly

Targets [T] : Generate targets.

- endless loop, except if received the terminate signal
- long period
- generate new coordinates for the targets with random values from /dev/urandom
- send the coordinates to the blackboard
- terminate cleanly

Parameter file [Config.txt] : Store the parameters values for the simulation.

3. list of components, directories, files

One directory : "Assignement1" with all the files :

- texte file : Config.txt
- code file : Init.c ; B.c ; D.c ; I.c ; O.c ; T.c
- make file : makefile

4. instructions for installing and running

The project requires the following components to compile and run correctly:

System Requirements

- A Unix/Linux operating system
- Access to the special device file /dev/urandom for random number generation
- POSIX-compliant system calls, including fork(), exec(), and wait()

Libraries

- ncurses (for terminal-based display and input) on Debian/Ubuntu systems, install with: "sudo apt install libncurses5-dev libncursesw5-dev"
- math library (math.h)
- Standard C library (stdio, stdlib, string, etc.)

Build Tools

- gcc
- make

5. operational instructions

Start program :

- make
- ./Init

Problem with window size :

Sometimes, if the program is started for the first time, it is possible that the konsole window is too small, and the print of the scene is bad. In this case, change the size of the konsole to the maximum, then close the program in the terminal with Ctrl+C and restart it.

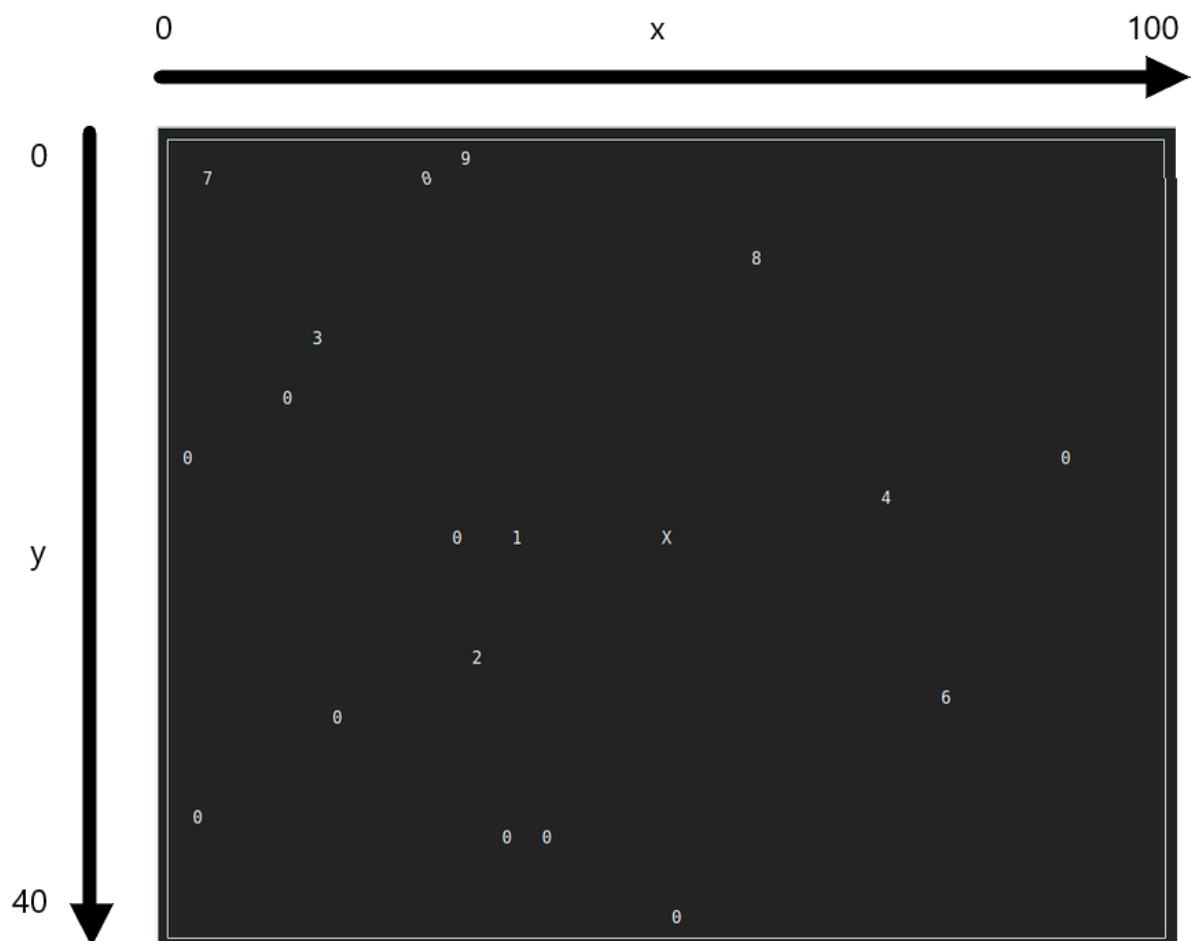
6. useful things

Protocol to communicate with the blackboard :

- The first message is always addressed to the blackboard. It indicates if the other program want to Read or write in the blackboard : 'r' or 'w'
- Then the blackboard answer :
 - the data if the program want to read
 - "ok" if the program wants to write, indicating the blackboard is blocked and is waiting for the values.
- The program then read or write the values

Each program asks to read / asks to write different variables, the buffers are very different between the pipes.

Scene coordinates :

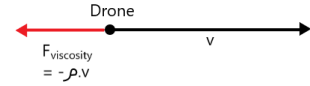


Dynamic calculation of the new coordinates :

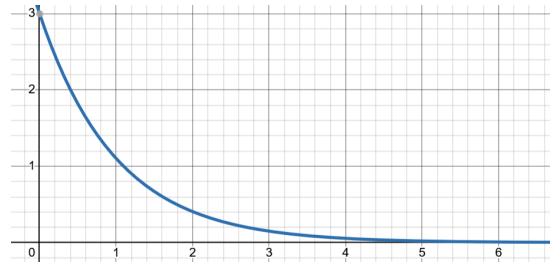
2nd Newton law : $\sum F = m \cdot a$ so $a = \frac{1}{m} \sum F$

$$\sum F = F_{motor} + F_{viscosity} + F_{obstacles}$$

F_{motor} is determined by the keys ; $F_{viscosity} = -\rho \cdot v$



$$F_{obstacle} = F \cdot e^{-\frac{distance}{repulsion\ radius}}$$



$$distance = \sqrt{(x_{drone} - x_{obstacle})^2 + (y_{drone} - y_{obstacle})^2}$$

$$F_x = F_{obstacle} \cdot \cos \theta$$

$$F_y = F_{obstacle} \cdot \sin \theta$$

$$\theta = \text{atan}((x_{drone} - x_{obstacle}) / (y_{drone} - y_{obstacle}))$$

